

Verification & Validation Plan - Revision 2

1. Verification Scope

- Registration correctness
- Scheduler execution
- Observation correctness
- Notification suppression and emission
- Restart persistence
- Concurrency safety

2. Positive Validation Scenarios

1. Registered task completes before first reminder.
2. Registered task remains active and receives periodic reminders.
3. Registered task fails and emits failure notification.
4. System restarts and still tracks active task.

3. Negative / Stress Scenarios

1. Duplicate registration attempt.
2. Task observation source disappears unexpectedly.
3. Scheduler fires while registration/update transaction is active.
4. Multiple tracked tasks exceed reminder interval simultaneously.
5. Notification emitter fails temporarily.

4. Evidence Requirements

Claim	Evidence
Task was registered	SQLite row + event row + command output
Scheduler executed	cron/systemd timer logs

Reminder sent	message/send evidence + DB event row
Duplicate notification suppressed	before/after DB state and absence of extra event row
Concurrency safety	contention test logs with no corruption or duplicate transition

5. Exit Criteria

- Positive completion/failure/reminder paths proven
- At least one negative test proven
- At least one concurrency/contention case proven
- At least one restart persistence case proven
- No false completion or false healthy notification in tested scenarios